Group theory question

M. Joris

1 Question

Let F be a free group, and let $S \subseteq F$ be a one-subset with $1 \notin S$. Put $K = \langle S \rangle$. Must there exist a surjective homomorphism

$$\phi : K \to \mathbb{Z}/3\mathbb{Z}$$

such that

$$S = \phi^{-1}(\{1, 2\})?$$

Equivalently, must $K \setminus S = \ker \phi$?

2 Motivation for the question

In the abelian case, every one-subset not containing the identity comes from a quotient map onto $\mathbb{Z}/3\mathbb{Z}$ it is the inverse image of the two nonzero classes. The same construction should work in any group (please double check this). If

$$\phi : K \to \mathbb{Z}/3\mathbb{Z}$$

is a homomorphism, then

$$S = \phi^{-1}(\{1, 2\})$$

is a one-subset. Indeed, if $x, y \in S$, then $\phi(x)$ and $\phi(y)$ are both nonzero in $\mathbb{Z}/3\mathbb{Z}$. Hence exactly one of $\phi(x) + \phi(y)$ and $\phi(x) - \phi(y)$ is nonzero in $\mathbb{Z}/3\mathbb{Z}$. Therefore exactly one of xy and xy^{-1} lies in S. Thus, for a one-subset $S \subseteq F$ with $1 \notin S$, the natural converse is to ask whether this is the only possible construction. If $K = \langle S \rangle$ and $K \setminus S$ is a normal subgroup of K, then the quotient $K/(K \setminus S)$ has exactly three elements, and the quotient map

$$K \to K/(K \setminus S) \cong \mathbb{Z}/3\mathbb{Z}$$

recovers S as the inverse image of the two nonzero classes. Therefore the question is asking whether every one-subset of a free group which does not contain the identity comes from a $\mathbb{Z}/3\mathbb{Z}$-valued character on the subgroup generated by S.